



Controllable Universal Fair Representation Learning

Yue Cui, Chen Ma, Kai Zheng, Lei Chen, and Xiaofang Zhou



Introduction

- Machine learning models developed for intelligent Web applications such as personalized search, online advertising and hiring platforms usually utilize user data with rich demographic features.
- Unfairness and biases that already exist in real-world data may be involuntarily introduced to these models.
- So, this paper proposed a method aiming to learn fair and transferable representations of users that can be used for a wide spectrum of downstream tasks

Problem

DEFINITION 2. (Attribute-wise) Demographic Parity (DP): [10] Given any learning algorithm $\mathcal{A}$, a sensitive attribute $\mathbf{c}_k$, $\mathcal{A}$ has demographic parity w.r.t. $\mathbf{c}_k$ if:

$$\Delta_{DP}(\mathcal{A}, \mathbf{c}_k) = \max_{i,j} |P(\hat{y} = 1 | \mathbf{c}_k = i) - P(\hat{y} = 1 | \mathbf{c}_k = j)| = 0$$

DEFINITION 3. Universal Demographic Parity (UDP): Given the collection of all sensitive attributes, i.e., $\mathbf{C}^U$, an algorithm follows universal demographic parity w.r.t. $\mathbf{C}^U$ if:

$$\Delta_{DP}(\mathcal{A}, \mathbf{C}^U) = \|\mathbf{v}_{\Delta_{DP}}\|_2 = \sqrt{\sum_{i=1}^{|\mathbf{C}^U|} \Delta_{DP}(\mathcal{A}, \mathbf{c}_i)^2} = 0$$

where $|\mathbf{C}^U|$ denotes the number of variables in $\mathbf{C}^U$, $\mathbf{v}_{\Delta_{DP}} = [\Delta_{DP}(\mathcal{A}, \mathbf{c}_1), \dots, \Delta_{DP}(\mathcal{A}, \mathbf{c}_{|\mathbf{C}^U|})] \in \mathbb{R}^{|\mathbf{C}^U|}$ is a vector formed by attribute-wise Δ_{DP} , and $\|\cdot\|_2$ is the L2 norm.

Table 1: Summary of Notations.

Notations	Descriptions
$\mathbf{c}_i, n_i$	A sensitive attribute and its total number of categories
$\mathbf{C}^l$	Collection of level l combined sensitive attributes
$\mathbf{C}^U$	Collection of the universal set of sensitive attributes
$\mathcal{A}$	A learning algorithm
$\Delta_{DP}(\mathcal{A}, \mathbf{c}_k)$	Demographic disparity on $\mathcal{A}$, sensitive attribute $\mathbf{c}_k$
$\Delta_{DP}(\mathcal{A}, \mathbf{C}^U)$	Universal demographic disparity on $\mathcal{A}$, universal sensitive attribute set $\mathbf{C}$
$\mathbf{x}, \hat{y}, y$	Input features, predicted outcome, and groundtruth label
$\mathbf{z}$	Learned representation vector
$I(p; q)$	Mutual information between distributions p and q
β	Tune-able parameter to control the trade-off between fairness and accuracy



Method - Objective Deriving

- Two goals
 - maximizing data utility
 - it requires the representation z to maximize the mutual information between itself and y
 - maximizing $I(y; z)$ equivalent to minimizing $-I(y; z)$
 - minimizing disparity
- mutual information has been proven to be promising to serve as a link between input features, intermediate representations and demographic parity.
- Two goals
 - Maximizing data utility
 - Maximizing the mutual information between z and y
 - Minimizing disparity
 - Minimizing the mutual information between z and c

only need to focus on level-1 sensitive attributes. Based on Theorem 5, we can rewrite the objective as follows

$$\mathcal{O} \triangleq \min_{p(z|x) \in \Omega} [-I(y; z) + \beta \sum_{i=1}^{|C^1|} I(z; c_i)] \quad (3)$$

where C^1 is the collection of all level-1 combined sensitive attributes, *i.e.*, the singular sensitive attributes.

Method

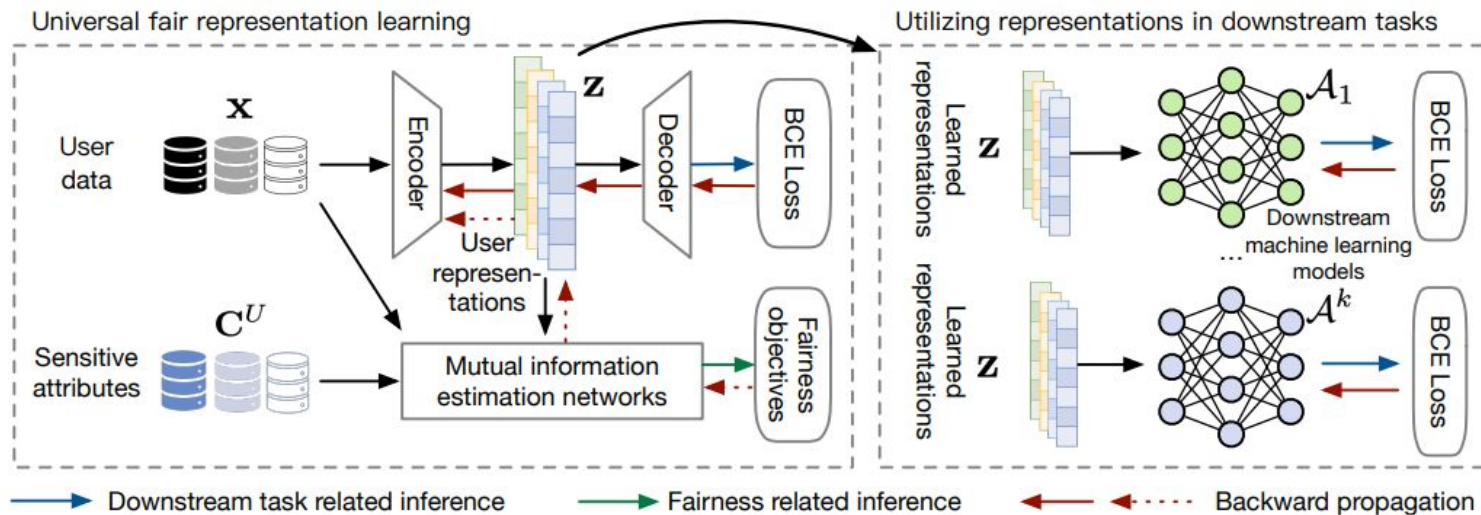


Figure 1: Overview of CUFRL.



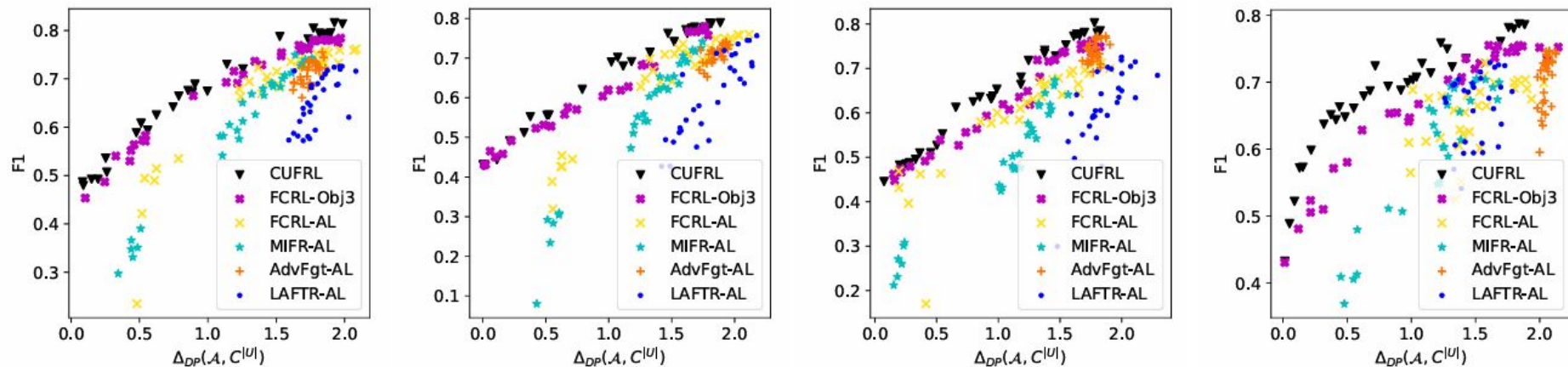
Method - Mutual information estimation

- Estimate $I(y; z)$
 - Encoder
 - Decoder
 - 1-layer MLP
- Estimate $\sum_{i=1}^{|\mathcal{C}^1|} I(\mathbf{z}; \mathbf{x} | \mathbf{c}_i)$.
- Loss function K instances from $\{\mathcal{B}_{joint}, 1\}$ and $\{\mathcal{B}_{prod}, 0\}$ separately, the loss function can be formulated as:

$$L_{c_i} = -\frac{1}{2K} \left[\sum_{(z, x, c_i) \in \mathcal{B}_{joint}} \log \omega(z, x, c_i) + \sum_{(z, x, c_i) \in \mathcal{B}_{prod}} \log(1 - \omega(z, x, c_i)) \right] \quad (9)$$

where $\omega(z, x, c_i)$ is the prediction of NN_D w.r.t. the probability of whether the sample is from the joint distribution. In other words, the

Results



(a) Logistic Regression - AdultCensus

(b) Random Forest - AdultCensus

(c) MLP-AdultCensus

(d) AdaBoost-AdultCensus

Figure 2: Trade-off between F1 and $\Delta_{DP}(\mathcal{A}, C^U)$ on the AdultCensus dataset.



Implementation

- Pseudo code

Algorithm 1 Compute L_c

- 1: **Input:**
 - 2: /* Assume representation z has been obtained in previous steps.*/
Batch data $\mathcal{B} = (z, \mathbf{x}, c)$, inner iteration T ;
 - 3: **Output:** L_c ;
 - 4: **for** $t \in \{1, \dots, T\}$ **do**
 - 5: Randomly split $\mathcal{B}$ equally to two parts $\mathcal{B}_{joint} = \{z_j, x_j, c_j\}_{j=0}^{n/2}$ and $\mathcal{B}_{gen} = \{z_j, x_j, c_j\}_{j=n/2}^n$;
 - 6: Compute $z'_j = NN_G(c_j) \in \mathbb{R}^{|C_1|}$, $\forall c_j \in \mathcal{B}_{joint}$ and obtain $\mathcal{B}_{prod} = \{z'_j, x_j, c_j\}_{j=0}^{n/2}$;
 - 7: Compute $L_{c_i, t}$ with $\{\mathcal{B}_{prod}, 1\}, \{\mathcal{B}_{prod}, 0\}$
 - 8: Monte Carlo average $L_c = \sum_i Avg_t(L_{c_i, t})$
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